

Bayesian Modeling of Diabetes Risk: Posterior Insights from BRFSS Health Indicators

1. Abstract

Diabetes remains a major global health challenge, affecting millions and leading to severe health complications if left unmanaged. This study analyzes key risk factors associated with diabetes using Bayesian modeling techniques. We utilized survey data from the Behavioral Risk Factor Surveillance System (BRFSS) and applied Bayesian logistic regression models, including Monte Carlo Markov Chain (MCMC) and variational inference methods, to estimate associations between lifestyle factors and diabetes risk. While we primarily focused on risk factors identified by the CDC, posterior predictive checks suggested that additional variables from the dataset could further improve predictive accuracy. Our results indicated that smoking and age were positively associated with diabetes risk, while fruit and vegetable consumption and regular physical activity were associated with lower risk. These findings highlight the importance of targeted public health interventions aimed at promoting healthy diets and exercise to mitigate diabetes risk.

2. Introduction

Diabetes is a chronic disease that affects both quality of life and long-term health outcomes for millions worldwide. Given its high prevalence and the burden it places on healthcare systems, understanding what factors contribute to diabetes risk is critical. In this paper, we model the probability of having diabetes based on various risk factors, using both Markov Chain Monte Carlo (MCMC) methods and variational inference techniques to estimate our model parameters. We perform posterior predictive checking as our method of checking the robustness of the model and interpret results given by our MCMC model.

3. Background

According to the International Diabetes Federation (IDF), around 463 million people worldwide are currently living with diabetes, and this number is expected to increase to 578 million by 2030. In the United States alone, approximately 38.4 million individuals have been diagnosed with diabetes, leading to about 16.9 million hospital visits where diabetes is recorded as a diagnosis. Treating diabetes also places a significant financial burden on the healthcare system, accounting for one in every four health care dollars spent in the U.S.

Given the widespread and serious effects of diabetes on health and health care costs, our group is interested in exploring the risk factors associated with the disease. In particular, we want to investigate: *Given a person's risk factors, what is their probability of having diabetes, and which factors play the most important role?* Through this project, we hope to better understand

what contributes to diabetes risk and highlight areas where prevention efforts may be most effective.

4. Data

4.1 Source

The data used in this project is published on Kaggle by Abdelaziz Sami and is a cleaned and consolidated version of the CDC's 2015 Behavioral Risk Factor Surveillance System (BRFSS) survey. The BRFSS is conducted through phone interviews, where participants report on their health status, behaviors, and medical history. It employs a complex sampling design that includes stratification and weighting to ensure that the collected data is representative of the U.S. population.

The specific dataset we use is titled *"diabetes_binary_5050split_health_indicators_BRFSS2015.csv"*, which is balanced for modeling purposes to create an equal split between individuals with and without diabetes. It contains 70,693 responses across 21 variables.

4.2 Variables

Based on the CDC article *"Diabetes Risk Factors,"* we identify several well-known contributors to diabetes risk, including family health history, age, weight, physical activity, and diet [3]. Drawing from these findings, we choose to focus on a subset of related variables available in our dataset and test their association with diabetes.

Specifically, we are interested in six predictors: "Smoke," "PhysActivity," "Fruits," "Veggies," "BMI," and "Age." The variable "Smoke" indicates whether a participant has ever smoked at least 100 cigarettes in their lifetime. "PhysActivity" records whether a participant engaged in any physical activity or exercise, excluding their regular job, within the past 30 days. "Fruits" measures whether a participant consumes fruits one or more times per day, and "Veggies" measures whether a participant consumes vegetables one or more times per day. "BMI" represents the participant's body mass index, calculated from self-reported height and weight, and "Age" records the participant's age.

4.3 Limitations of the dataset

Although the dataset is comprehensive, there are some limitations that could affect the reliability of our findings. One important limitation is that the dataset does not distinguish between Type 1 and Type 2 diabetes. Type 1 diabetes is caused by an autoimmune attack on the insulin-producing cells in the pancreas and usually develops during childhood or adolescence, while Type 2 diabetes results from insulin resistance or insufficient insulin production and tends to develop in adulthood. Because these two types of diabetes have different causes and age patterns, not separating them could make age a less reliable predictor in our analysis.

Another limitation is that the data is entirely self-reported. This could introduce bias in several ways. Participants might misunderstand questions, leading to misinterpretation. Memory

issues could also affect how accurately participants report past behaviors like smoking or physical activity. In addition, response bias, where participants answer in a way they think is more socially acceptable, could further reduce the accuracy of the data.

4.4 Potential improvement of the dataset

Future studies could improve reliability by distinguishing between Type 1 and Type 2 diabetes. Since Type 2 diabetes accounts for most cases and tends to occur later in life, focusing only on Type 2 could make age a stronger and more meaningful predictor.

It would also help to double-verify diabetes status through clinical records instead of relying only on self-reports. Using online or mobile surveys could reach a broader and more diverse population, and adding follow-up surveys would allow researchers to track changes over time. Finally, applying appropriate weighting methods would help adjust for the survey's complex sampling design and make the results more generalizable.

5. Model

5.1 Notations

$y_i \sim \text{Bernoulli}(\pi_i)$, where y represents their diabetes status.

$\pi_i = \text{logit}^{-1}(\eta)$, where π_i represents the probability of person i having diabetes, and η is our linear predictor given by the equation below:

$$\eta = \beta_0 + \beta_1(\text{SmokingStatus}_i) + \beta_2(\text{BMI}_i) + \beta_3(\text{Age}_i) + \beta_4(\text{PhysActivityStatus}_i) + \beta_5(\text{FruitsStatus}_i) + \beta_6(\text{VeggieStatus}_i)$$

5.2 Priors

We used weakly informative priors for our Bayesian logistic regression model, following recommendations from Andrew Gelman and the Stan development team. Given our large dataset of approximately 70,000 observations, the data would largely dominate the posterior, but priors help prevent extreme coefficient estimates. For continuous predictors such as BMI and Age, and for binary status variables like SmokingStatus and PhysActivityStatus, we assigned $\sim \text{Normal}(0, 1)$ prior to moderately shrink estimates toward zero without overly restricting them. These choices provided regularization while still letting the data drive the results.

5.3 Likelihood

In this study, we calculated the theoretical likelihood for completeness.

Each response $y_i \in \{0, 1\}$ is modeled as

$$P(y_i | \pi_i) = \begin{cases} \pi_i, & y_i = 1, \\ 1 - \pi_i, & y_i = 0, \end{cases}$$

which can be written compactly as

$$P(y_i | \pi_i) = \pi_i^{y_i} (1 - \pi_i)^{1-y_i}.$$

We link the success probability π_i to a linear predictor η_i :

$$\eta_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip}, \quad \pi_i = \text{logit}^{-1}(\eta_i) = \frac{1}{1 + e^{-\eta_i}}.$$

Assuming independence across $i = 1, \dots, n$,

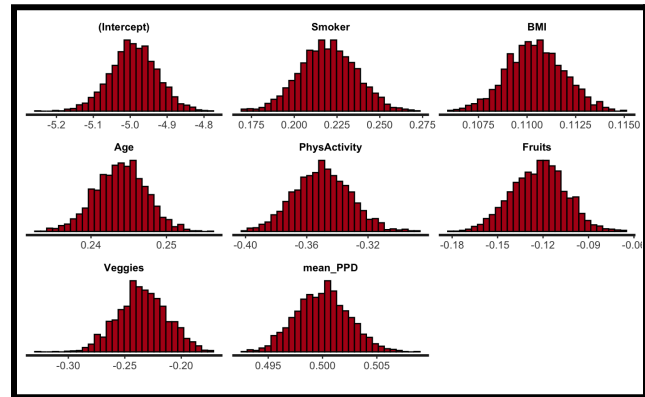
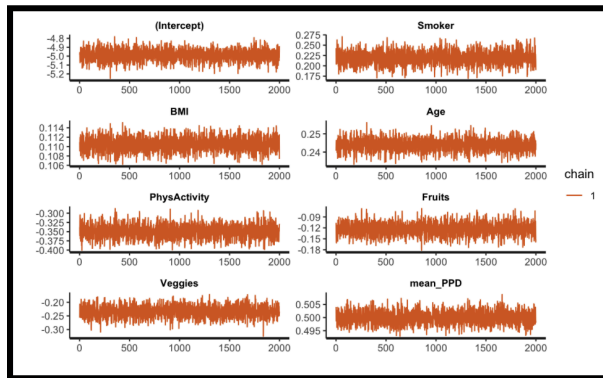
$$L(\boldsymbol{\beta} | \mathbf{y}) = \prod_{i=1}^n P(y_i | \pi_i) = \prod_{i=1}^n [\pi_i]^{y_i} [1 - \pi_i]^{1-y_i}.$$

5.4 Computational Strategies

Because our priors and likelihood are not conjugate models, we need to utilize sampling techniques in order to obtain our posterior distribution. We applied both Variational Inference (VI) and Markov Chain Monte Carlo (MCMC) methods to estimate the posterior distribution. Variational Inference was used for its computational efficiency given the large sample size (~70,000 observations), while MCMC provided more posterior accuracy. Posterior mean estimates and credible intervals were highly consistent across both methods, indicating that the variational approximation was sufficiently accurate for our model and inferential goals.

5.5 MCMC

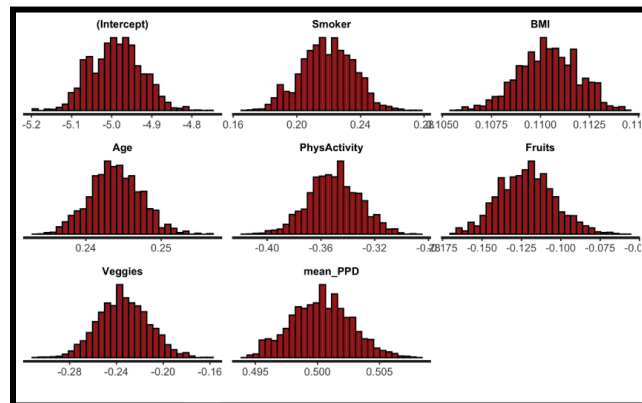
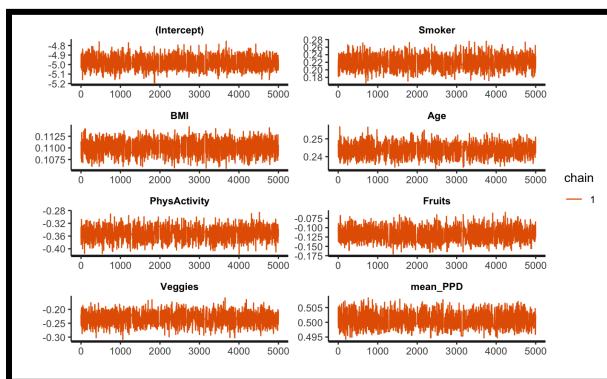
We ran Hamiltonian MCMC with 5,000 iterations, 1,000 warmups, and 4 chains. The table summarizes posterior means as well as 95 percent credible intervals, as well as effective sample sizes for each parameter. Each chain showed good convergence, with little pathologies found. Credible intervals excluded zero for all predictors, suggesting strong evidence that each variable does have an effect on log-odds of having diabetes.



5.6 Variational Inference

In addition to MCMC, we performed Variational Inference (VI) to approximate the posterior distributions. VI is a faster, optimization-based method that fits a simpler family of distributions (Gaussian) to the true posterior.

The results obtained through VI were highly consistent with those from MCMC, suggesting that the true posterior was approximately Gaussian and well-behaved in this analysis. However, for final inference, we relied primarily on the MCMC results, which provide a more accurate and less biased characterization of posterior uncertainty.

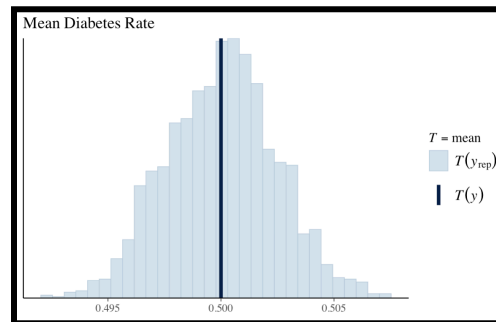


5.7 VF and MCMC Comparisons

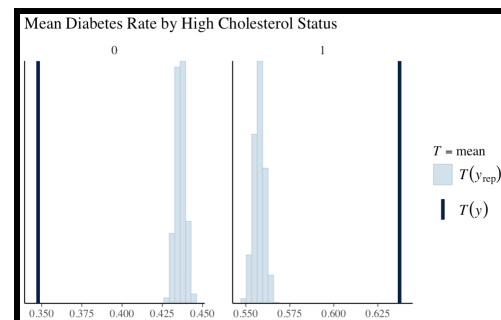
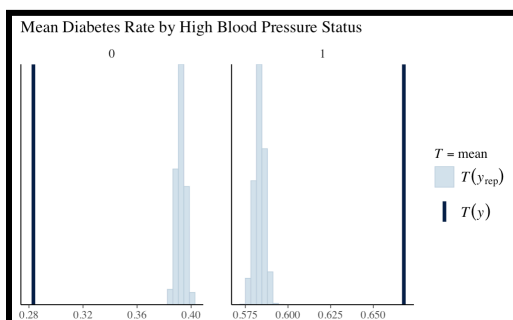
Compared to MCMC, Variational Inference produced similar posterior means and standard deviations across all parameters, indicating that the posterior distributions were approximately Gaussian. However, VI tended to slightly underestimate posterior uncertainty (narrower credible intervals), as is common due to its optimization-based approximation. Given this, we relied on MCMC for final inference to ensure a more accurate representation of posterior variability.

6. Model Checking

Once we've constructed a probability model and computed the posterior distribution of all estimands, the next thing to do is assess the fit of the model to the data since Bayesian inferences assume the whole structure of a probability model and can yield misleading inferences when the modeling is poor. Therefore we use posterior predictive checking to see if replicated data generated under the model looks similar to the observed data. The plot below indicates that the posterior predictive samples are consistent with the original dataset, which is the line we see at 0.5.

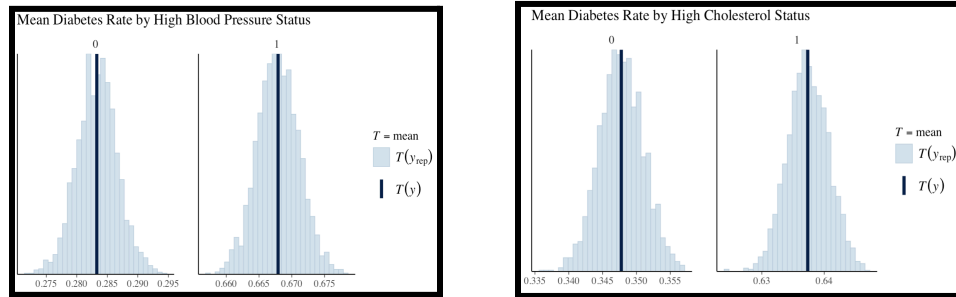


However, note that only five out of twenty one variables were included in the model. These five were selected according to details found on the CDC website [3] and so some crucial information may have been lost by omitting the other 16 variables. Further research seems to indicate that high blood pressure and high cholesterol also have high correlation with diabetes [1,2]. We use predictive checking using these omitted variables to check the robustness of our results. Posterior predictive samples are used to compute the proportion of diabetes among those with high blood pressure and those without, and we see that results shown below are not consistent with the proportions observed in the original dataset. We see similar results among those with high cholesterol and those without.



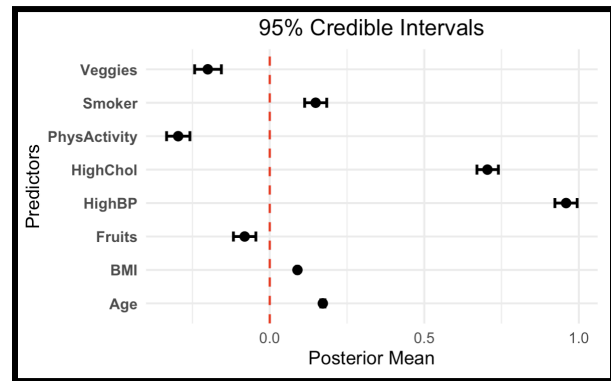
To address this issue discovered in the posterior predictive check, a new model is fitted that includes high BP and high CHOL. Computing posterior predictive samples using the new model yields more robust results as seen down below.

However, there may be other statistically significant variables that may have evaded our notice. Rather than performing posterior predictive checks for every single variable, a more efficient step might be to change the way of performing variable selection. One approach to consider is the Lasso Regression model. This would entail fitting a Lasso using all the variables, and using the lambda value selected through Cross-Validation, select the variables that have not been shrunk to zero.



7. Results

We leave the idea of variable selection as motivation for future work. For now, we interpret results for our final model after having included HighChol and HighBP in our predictors. Since the credible intervals do not contain zero, this indicates evidence that all listed predictors are associated with diabetes risk. A brief overview of the plot shows that variables such as veggies, physical activities, and fruits are negatively associated with diabetes, while smoking, high cholesterol, high blood pressure, bmi, and age showcase a positive correlation with diabetes.



parameter	Mean	SD	2.5% CI	97.5% CI
(Intercept)	-4.739	0.066	-4.868	-4.606
Smoker	0.149	0.018	0.113	0.184
BMI	0.090	0.002	0.087	0.093
Age	0.172	0.003	0.165	0.179
PhysActivity	-0.296	0.020	-0.334	-0.258
Fruits	-0.081	0.018	-0.117	-0.045
Veggies	-0.200	0.022	-0.243	-0.156
HighBP	0.959	0.018	0.923	0.995
HighChol	0.704	0.018	0.670	0.740
mean_PPD	0.500	0.002	0.495	0.504

The table here provides further numerical interpretations of the plot given above. We observe that having high blood pressure is associated with an increase in log-odds probability of getting diabetes by approximately 0.96 holding all other factors constant. Similarly, having high cholesterol is correlated with a 0.704 in log-odds probability of getting diabetes. While

high cholesterol and high blood pressure show the strongest positive associations, we also find that age, bmi, and smoke have a positive correlation with approximate increases in log-odds of 0.172, 0.09, 0.149 respectively.

On the other hand, performing physical activities, and eating fruits or vegetables at least once per day is negatively associated with the risk of diabetes, with a decrease in log-odds probability of getting diabetes by approximately 0.296, 0.081, and 0.2 respectively.

8. Conclusion/Future

Results indicate that a high BMI, being older, being a smoker, having high blood pressure, and having high cholesterol are all positively associated with the risk of diabetes. On the other hand, performing physical activities and eating veggies/fruits is negatively correlated with the risk of diabetes. Our Hamiltonian MCMC model has provided results that align with our intuition, and our posterior predictive checking has further supplemented the robustness of the sample results. However, as indicated above, the lack of distinction between Type 1 and Type 2 diabetes may affect the reliability of some predictors such as age and HighChol, since both have a stronger association with Type 2 diabetes [2]. Another limitation in our dataset is that the data is self-reported and can lead to errors through misinterpretation, memory issues or bias. We have also addressed the fact that variables were selected based on research, which is not a robust method and critical information may have been omitted in the process.

Therefore, the following steps are to be considered to improve on these shortcomings to provide a more robust model with reliable results. In order to improve the quality of the dataset, it may be necessary to validate diabetes status through clinical records and improve survey design. Furthermore, distinguishing between Type 1 and Type 2 diabetes and fitting the model using either as the response variable may lead to more reliable results. Finally, we propose using a lasso regression model to perform variable selection, which will yield more significant results since we are taking into account all 21 variables in the dataset.

Sources

- [1] Johns Hopkins Medicine. (n.d.). *Diabetes and high blood pressure*. Johns Hopkins Medicine. <https://www.hopkinsmedicine.org/health/conditions-and-diseases/diabetes/diabetes-and-high-blood-pressure>
- [2] American Heart Association. (n.d.). *Cholesterol abnormalities & diabetes*. <https://www.heart.org/en/health-topics/diabetes/diabetes-complications-and-risks/cholesterol-abnormalities--diabetes>
- [3] Centers for Disease Control and Prevention. (2022, June 30). *Diabetes risk factors*. U.S. Department of Health and Human Services. <https://www.cdc.gov/diabetes/risk-factors/index.html>